Aman Shaikh F - 02

Aim: To study and implement feature selection techniques using:

- Filter Method (Correlation-based Feature Selection)
- Wrapper Method (Recursive Feature Elimination – RFE)

```
import pandas as pd
import numpy as np
from sklearn.feature_selection import RFE
from sklearn.linear_model import LogisticRegression
```

```
df = pd.read_csv("/content/housing.csv")
X = df.drop("median_house_value", axis = 1)
y = df["median_house_value"]
print(X.shape)
```

```
(20640, 9)
```

```
df = X.copy()
```

```
df['median_house_value'] = y
df_numeric = df.select_dtypes(include = 'number')
target_corr = df_numeric.corr()['median_house_value'].drop('median_house_value')
print("\nCorrelation with Target:")
print(target_corr.sort_values(ascending = False))
selected_features = target_corr[abs(target_corr) > 0.3]
print(selected_features)
```

```
Correlation with Target:
median_income         0.688075
total_rooms           0.134153
housing_median_age    0.105623
households            0.065843
total_bedrooms        0.049686
population           -0.024650
longitude            -0.045967
latitude             -0.144160
Name: median_house_value, dtype: float64
median_income    0.688075
Name: median_house_value, dtype: float64
```

```
from sklearn.linear_model import LinearRegression

X_numeric = X.select_dtypes(include='number').fillna(X.select_dtypes(include='number').median())
y_numeric = y

model = LinearRegression()
rfe = RFE(estimator = model, n_features_to_select = 5)
rfe.fit(X_numeric, y_numeric)
print("\nSelected Features")
print(X_numeric.columns[rfe.support_])
```

```
Selected Features
Index(['longitude', 'latitude', 'housing_median_age', 'total_bedrooms',
       'median_income'],
      dtype='object')
```